WEEDOUT: A SEMI-SUPERVISED NON-CNN CLASSIFICATION BASED A-CPS FOR WEED MANAGEMENT

A PREPRINT

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ABSTRACT

The recent integration of Internet-of-Agro-Things (IoAT) and Artificial Intelligence (AI) in Smart Agriculture helped in maximizing yields within constrained resources. Similarly, since weed infestation causes a reduction in yield and quality of the produce many automation solutions were developed for managing weeds. Most of the methods try to detect /classify weeds by their visual features using CNN. But, different geographic, and climatic conditions across the globe and various stages in the growth cycle of the plant will cause similar crops from different places and stages to look different. Further, CNN models need a lot of images to train and perform poorly in classifying the images they had not been trained on. These CNN models can't be used in cases of unknown weed presence. This paper, WeedOut presents an Agriculture Cyber-Physical System (A-CPS) for comprehensive weed management which utilizes the shape and size of the crops to classify them instead of visual-feature dependent CNN methods. It is unaffected by any feature varying conditions and does not need any prior training. WeedOut clusters crops with similar shapes, sizes and classifies them by domain knowledge of the farmer in a semi-supervised approach with minimal human intervention. The proposed method achieved an accuracy of 89% when tested on generated images of farmland using images of individual crops.

Keywords Smart Agriculture; Agriculture Cyber-Physical System (A-CPS); Internet-of-Agro-Things (IoAT); Artificial Intelligence; Computer Vision; Semi-Supervised Learning; Weed Pressure; Weed management.

1 Introduction

Changes in climatic conditions, weather patterns and lack of awareness have resulted in depletion of natural resources. In addition, rapid growth in population and urbanization reduced the extent of land suitable for agricultural practices. With these given conditions, traditional agriculture was not able to produce enough food to serve the growing needs of the population. So, various innovative methods were introduced in agriculture with the help of advancements in computer science and instrumentation leading to a new paradigm called "Smart Agriculture" [1]. Various agricultural routines were automated by explicit coding for hand-crafted rules using Internet-of-Agro-Things (IoAT) as part of smart agriculture [2]. The development of Artificial Intelligence (AI) and Machine Learning (ML) introduced ways for training machines with the help of historical data and tackling future situations without being explicitly programmed. Implementations of these helped farmers automate agricultural routines with much precision for "Precision Farming" [3]. Any automation needs a couple of sensors to get information or data from physical entities and some computational resources to analyze the data and present results to farmers. Similarly, agriculture process automation needed the fusion of IoAT devices into agriculture infrastructure. Such agriculture infrastructure assigned with specific tasks like seed sowing, irrigation and harvesting is referred to as Agriculture Cyber-Physical System (A-CPS). This paper presents a comprehensive view of an A-CPS designed for weed management in agriculture.

Along with the primary crop in the farmland, some unwanted crops grow naturally without the need for farmer intervention. These unwanted plants, known as weeds, compete with the primary crop for essential resources such as water, soil nutrients, sunlight, and space. This competition can significantly affect the growth and yield of the primary crop. These weeds also act as a habitat for various disease-causing pests and fungi causing the spread of diseases in farmland. Furthermore, the presence of weeds causes soil erosion. To preserve the quality and quantity of produce

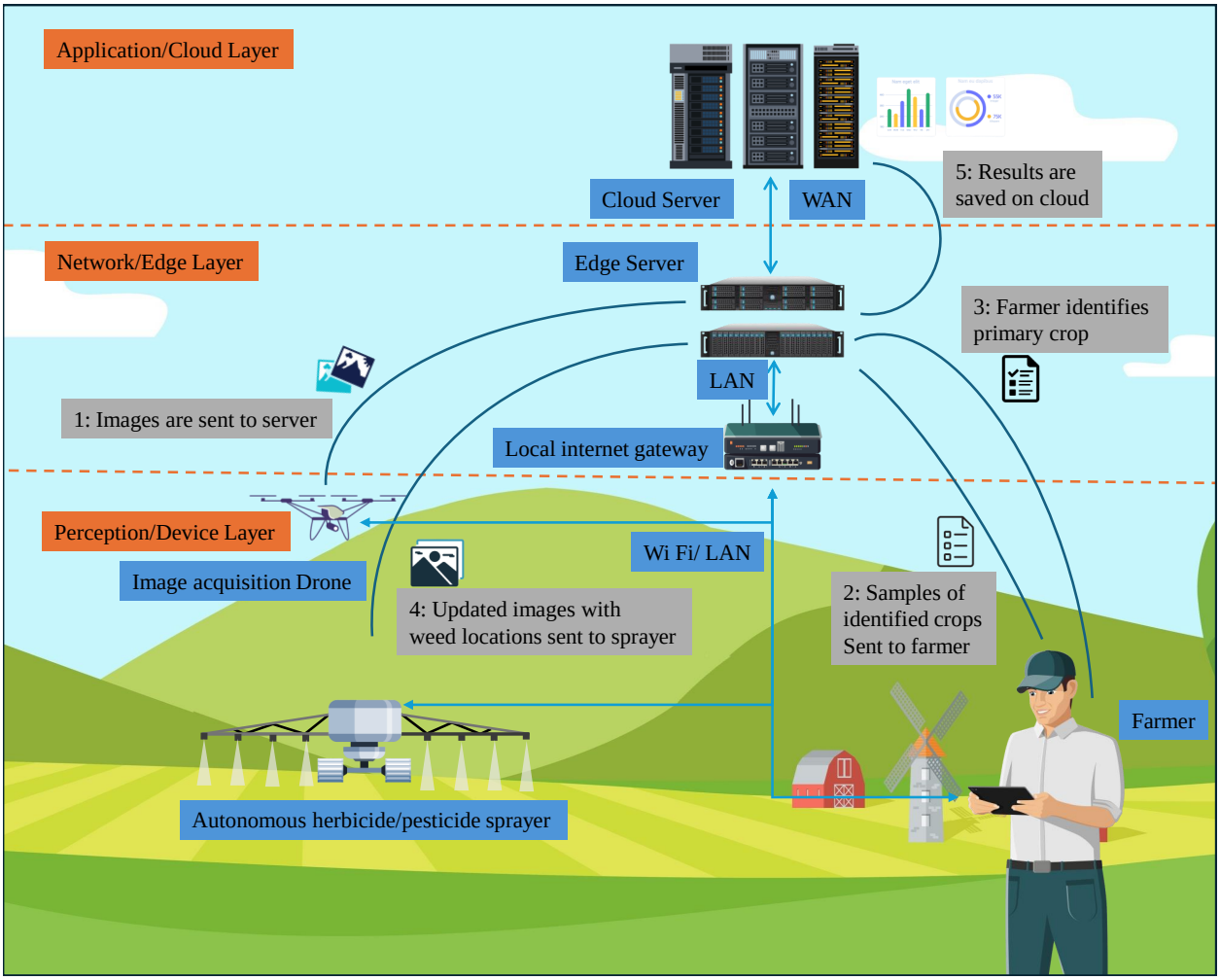


Figure 1: Overview of proposed Agriculture Cuber-Physical System (A-CPS) built using Internet-of-Agro-Things (IoAT).

in agriculture [4], it is essential to identify the presence of weeds and eliminate them. There are various methods for controlling weeds such as the traditional process of physically removing the weeds, the usage of herbicides to kill or suppress the growth of weeds, practicing methods like crop rotation and inter-cropping, using insects or fungi that feed on weeds. Among these methods, spraying of herbicide (chemical management) is the most widely preferred.

Spraying weeds with herbicide can be done by humans manually in cases where the farmland is considerably small. But if the farmland is spread over hundreds of acres, manual spraying requires a significant amount of time and manpower making it practically not feasible. Also, inappropriate usage of herbicides can have several negative effects on the environment [5]. Spraying herbicide on bare soil affects soil fertility and microbial activity impacting plant growth. When such a farm is watered, the herbicide residue is washed out and gets mixed up with local water bodies, contaminating them. Overdosage of weeds may accelerate the evolution of herbicide-resistant weeds and can also cause respiratory issues, skin diseases in humans exposed to them. To effectively use the right amount of herbicide while reducing manpower, the A-CPS should be able to navigate across the farmland and identify the presence of weeds along with their locations. So that, these findings can be used by autonomous weed sprayers [6] to spray herbicide at locations of weeds to eliminate them.

The current article introduces WeedOut, an A-CPS designed for weed management. WeedOut implements a non-CNN semi-supervised approach, as depicted in Figure 1. It leverages the shape of crops and the knowledge of farmers to identify, label various crops present in images of farmland as primary crops and weeds. Those annotated images are color-coded so that the sprayer can identify locations of weeds and spray herbicide at the locations of weeds.

Following this introduction, the remainder of the paper is organized as follows: Novel contributions of this article presented in Section 2 are followed by a brief discussion on prior works related to this article in Section 3. Proposed solution is explained in detail in Section 4 while experimental results are discussed and analyzed in Section 5. Section 6 summarizes and concludes the article providing insights into future direction.

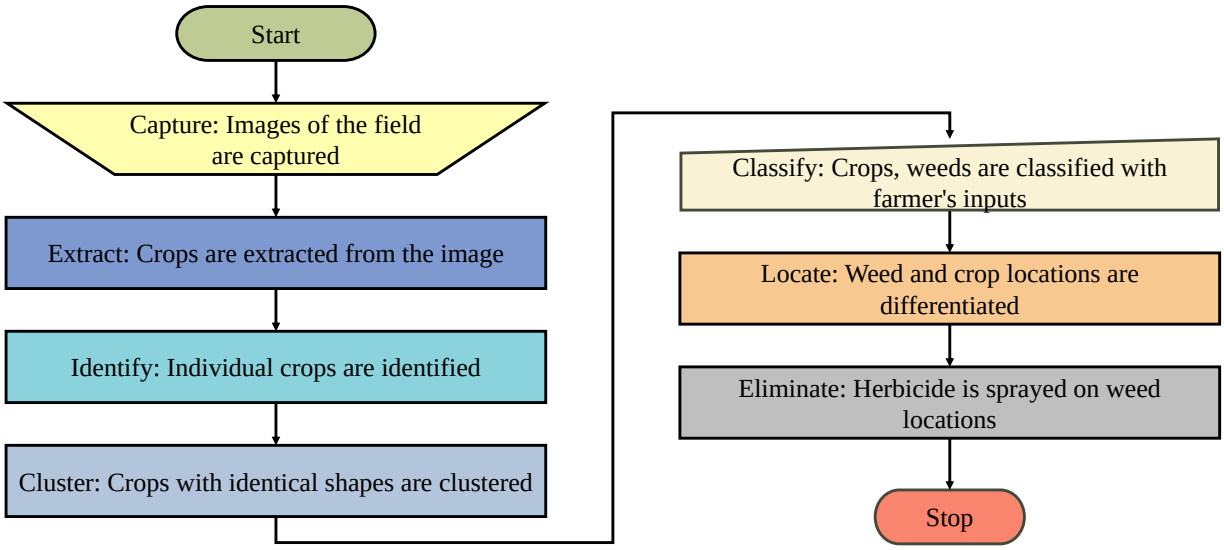


Figure 2: Workflow of WeedOut.

2 Novel Contributions of the Current Paper

2.1 Problem Statement

There have been many solutions to detect weeds using Convolution Neural Networks (CNN) like [7], [8] using R-CNN, [9] using YOLO , [10] using ResNet, AlexNet in [11], VGG in [12] with high accuracy and efficiency. CNN models use a sliding window technique where the image is scanned by numerous filters to detect visual features of the image in available color channels. When a set of features is found in an image, the image will be classified correspondingly. Training a CNN model involves curating a large number of such masks in various sizes so that the model will be able to capture and distinguish the spatial and visual features related to each class. If it is supposed to detect a car, it looks for car-shaped edges with a set of wheels in spatial proximity to a set of windows inside the edge. In the case of classifying crops, they will have to be trained to identify the texture, color, edges, shape, other visual features of the leaf and spatial features like the distance between leaves, alignment of the branches. To be able to train an efficient CNN model, it needs many images of crops that exhibit their features distinctly and are accurately labeled by humans. However, labeling of images involves a significant amount of input and time from humans. Even if the model is trained to demonstrate higher accuracy, those models will not be able to detect /classify an image accurately if the image belongs to a class that was not present in the training data since it was only able to fit masks that can detect features in the images presented for training.

Many environmental conditions like temperature, humidity, soil composition, nutrient levels, fertilizers used, natural genetic changes and the growth cycle of the crop show an impact on the physical appearance of a crop. In an effort to develop a CNN model to work universally, the labeled data needed becomes very large and obtaining all such images is impossible [13]. Also, there is a possibility be new unseen weeds being discovered. To develop a classification model to be able to work in most of cases and with minimal human efforts, geometric features like length and width of the leaf, shape of the crop, area occupied by each crop and any patterns in plantations can be instrumental.

2.2 Proposed Solution of the Current Paper

WeedOut presents an A-CPS in smart agriculture framework to identify and eliminate any weeds present in the farmland to protect primary crops. The proposed A-CPS adopts a three-tier IoT architecture with Application/Core layer, Network/Edge layer and Perception/Device layers as illustrated in Figure 1. Image acquisition systems and

autonomous weed sprayers work together with farmers in the loop to execute a sequence of processes shown in Figure 2 to manage weeds in the farmland.

Images of the farmland captured by drones are processed to extract the crop images by eliminating the soil background. To classify these crop images, instead of using visual-feature dependent CNN methods, WeedOut proposes the use of the shape and size of the crops (profile plots). Individual crops are identified from the crops in the images and crops with similar shape, size are grouped by unsupervised clustering. The farmer identifies the clusters with images of primary crops from the samples of each cluster presented to him. All the crops belonging to the clusters identified by the farmer are labeled as primary crops while the others are labeled as weeds. Thus, WeedOut classifies primary crops and weeds in a semi-supervised fashion with minimal inputs from the farmer, without the need for any prior training and reduces the effort of labeling many images.

With all the crops labeled as either weed or primary crop, weed pressure is computed and all the crops labeled as weed are colored in red and the rest are colored in green. These images will be saved on a cloud server for future reference and are presented to an autonomous weed sprayer. These Weed sprayer robots scan the presented images and differentiate the presence of weeds by their red color and spray herbicide at those locations of weeds to kill them. By targeted spraying, herbicide can be efficiently utilized and wastage can be reduced.

2.3 Novelty and Significance of the Proposed Solution

The novel contributions of this article are as follows.

1. The classification model presented in WeedOut is adaptable to environmental variations because it does not rely on hard rules that are dependent on the visual features of the crops. Instead, it leverages profile plots of the crops to classify them.
2. Dynamic Time Warping (DTW) method used to cluster similar crops based on similarity between the profile plots does not need any prior training. This eliminates the need for extensive human efforts associated with image annotations.
3. WeedOut tries to combine machine and human intelligence by incorporating human inputs through a semi-supervised method. This approach enhances the robustness, enabling context-aware classification of primary crops and weeds.
4. The Proposed method does not rely on computationally intensive methods like CNN, making it suitable to run on end devices like mobile phones and tablets. This scalability enables WeedOut to be used in any farmland.

3 Related Prior Works

Several non-CNN methods were developed to identify weeds in the farmland, a few of these recent methods will be discussed here. Authors of [14] use a multi-spectral imaging system to acquire images of farmland and apply the Fourier, Hough transform to detect rows of plantation and their orientation, identifying crop margins. Crops are classified by using spatial methods which consider crops outside of crop margins as weeds and a spectral method which classifies crops with lower NDVI at the center as weeds to attain better accuracy. [15] presented a population-based method that assumes that the number of weeds between crop margins is greater than the number of primary crops. The authors utilized depth data to identify crop rows and margins. All the crops outside of the crop margin are classified as weeds and the crops lying between crop margins are clustered into two clusters by KNN algorithm on their geometric features. The smaller cluster is classified as the primary crop and the larger cluster as a weed. In [16], crop lines are derived and super pixels of the image are obtained by SLIC algorithm. All the super pixels that are in contact with crop lines are considered primary crops and those with a 40°difference in orientation are classified as weeds. The leftover super pixels are classified by the label of their neighbors.

While the methods discussed above rely on row patterns in cultivation and seed sowing, the following methods rely on the area occupied by individual crops. Assuming that the area occupied by individual primary crops is greater than that of a weed crop, articles [17], [18] presented methods to calculate the number of pixels occupied by each crop as the area occupied by the crop and classified the crops whose area is above a predefined threshold as primary crops and the rest as weeds. Contrary to this assumption, authors of [19] and [20] assumed that the area occupied by the individual primary crop is lesser than that of a weed crop and classified crops with area below the predefined threshold as primary crops and others as weeds.

A model based on the shape of the crops which makes no assumptions but needs training data has been detailed in the article [21]. The authors proposed a method that compares the Active Shape Models (ASM) of the presented crop with the ones in its memory to classify the presented sample. A summary of these works is presented in Table **??**.

Differing from the aforementioned approaches, WeedOut neither relies on patterns in cultivation nor makes any assumptions related to the features of crops. Similar to [21], it utilizes shapes of the crops but with no prior training to group crops that are alike. All such identified groups are presented to the farmer so that he can identify which of them represents the primary crop.

4 Proposed Method - WeedOut

The proposed solution WeedOut, as described by the current article, is an A-CPS comprising of device/machines like drones for image acquisition, servers for processing the images, end devices like mobile phone/tablet to view results, autonomous spraying robots with a human (farmer) in the loop. Figure 3 outlines several stages for weed management in WeedOut, which are detailed in the following subsections.

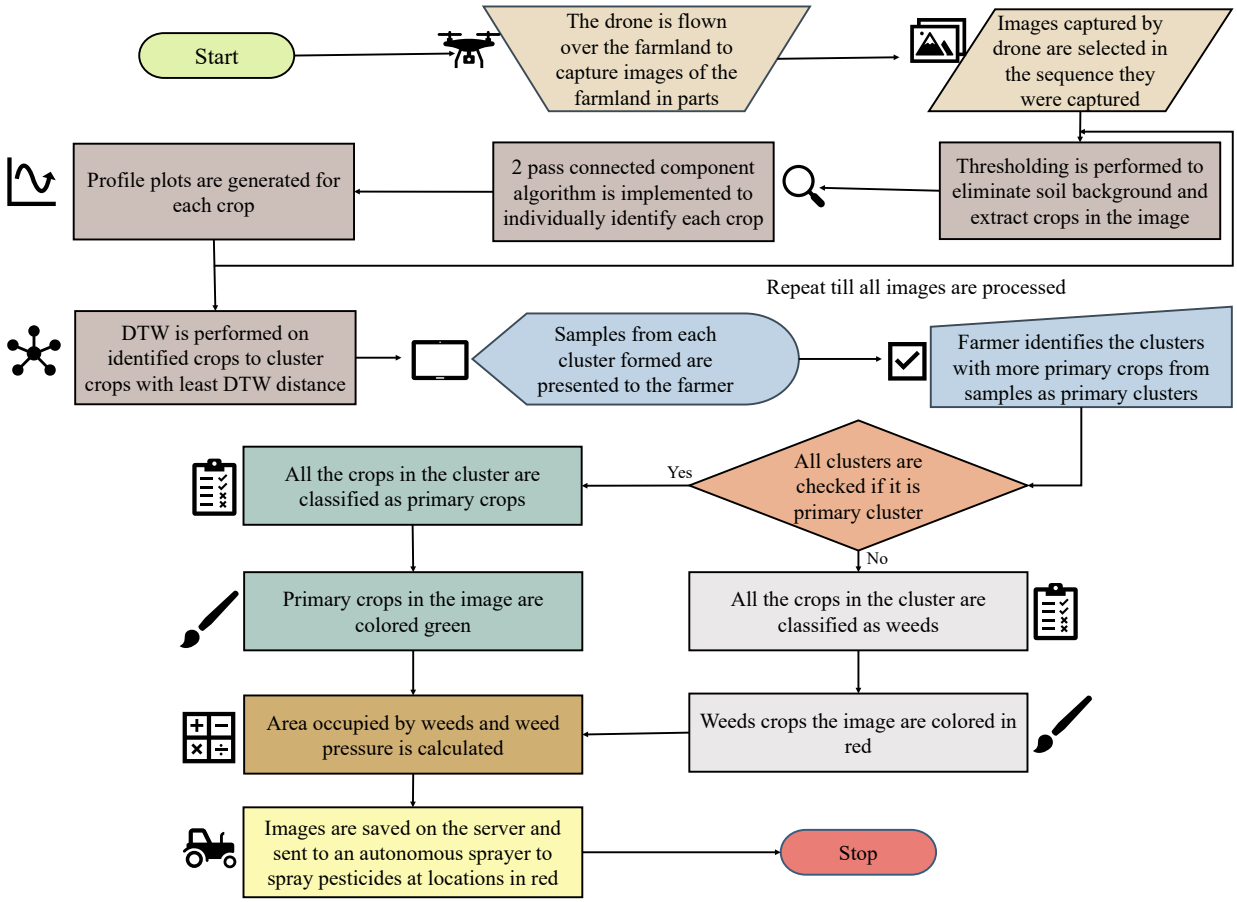
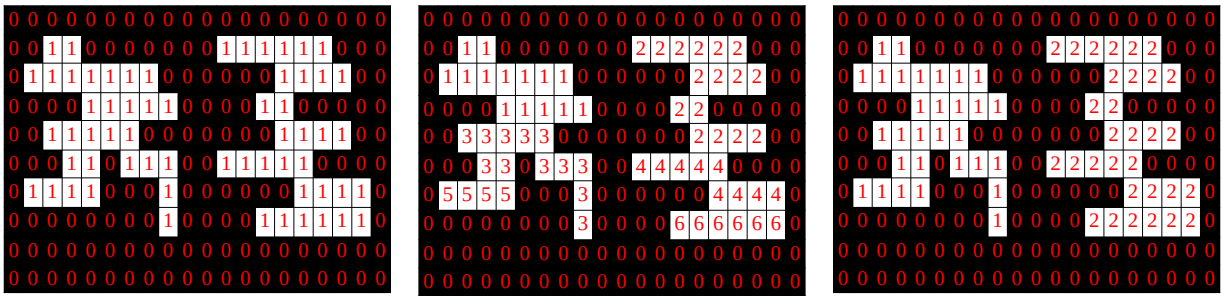


Figure 3: Operational flowchart of WeedOut.

4.1 Capture and Extract Blobs of Individual Crops in Images

To acquire images of the farmland a drone/rover equipped with a camera is flown across the entire farmland [22], [23]. It captures photos of the farmland in sequence so that when the photos are put together in the same sequence, they represent the farmland. The photos captured predominantly contain a soil background with crops in the foreground. The classification of weeds and primary crops is only related to the crops and spraying is performed on crops. To extract blobs of the crops from each image thresholding is performed on all the images. Since crops are in green color and the soil is brown all the time, limits for green color are defined in HUE color space. Then the image is also converted into HUE color space and all the pixels whose intensities are beyond defined limits are set to black [24], [25]. As a result, we only have the green-colored crops on a black background in the image eliminating the soil background. The updated images are resized to 450×300 so that all the images are of the same size and produce comparable



(a) Binary image (b) First pass (c) Second pass

Figure 4: Demonstration of two Pass Connected Component algorithm.

results. 3-channel RGB image is then processed to generate a single-channel binary image where "0" represents the background and "1" represents the crop in the image as in Figure 4. Thus, images of farmland have been converted into a 450×300 binary matrix.

4.2 Identify Individual Crops in Images

While capturing images of the farmland, a rover/drone tries to capture images of parts of the farmland rather than capturing each crop. So, there is a high probability that there could exist more than one crop in the images captured. To efficiently classify each crop, from the images captured each crop has to be identified uniquely and classified. The binary image generated in the prior sub-process is passed through a two-pass connected component algorithm [26], which is a graph-based image processing algorithm to label connected components in the binary image by passing through the image twice. In the first pass, the algorithm reads the values of each pixel (each element in the binary matrix), when a bright pixel (element with value 1) is encountered it is considered an object. Then, all the immediate neighbors of the object pixel are examined if they have any label assigned. If there is a label assigned to its neighbors, the highest of the labels is assigned to the object pixel. If not, a new label is assigned to the pixel starting from integer value 1 in ascending order. By the end of the first pass, every pixel has a label assigned to it (elements in the matrix are replaced by their labels).

In some cases, there can be forks/branches in the object. Since the algorithm assigns a new label when there are no neighbors around, different forks of the same object may be assigned with different labels. Due to this, pixels belonging to the same object can have multiple labels as in Figure 4. During the second pass, all the differences in labels are captured by saving equivalence between labels that share boundaries in an equivalence table. Then, all the labels that belong to a single object are identified and merged into a single label using the Union-Find algorithm. As a result, the binary matrix is converted to a matrix of positive integers where the background is identified by 0 and each crop component is identified by an integer value in just 2 passes. The number of labels generated ranges from 1 and the total number of objects present in the image as depicted by Figure 4. A detailed algorithm of two-pass connected component labeling is presented in Algorithm **??**.

4.3 Cluster Identical Crops in the image

Every plant species exhibits individual geometric features like length, width of leaves and orientation of leaves which predominantly define how a plant looks as a whole. Therefore, the change in width of the plant along with length when seen from the top of the crop and the area the crop can occupy also vary from species to species. The plot which shows the width of the crop along its length from one end to the other is considered a Profile Plot of a plant. Figure 5 shows profile plots of three different crops demonstrating how profile plots can help differentiate crops. Profile plots for each crop can be generated from the resultant matrix of the two-pass connected component algorithm. Since each label in the matrix represents an individual crop, a profile plot of each crop can be generated by counting the instances (columns) of the label in each row for all the rows where the label exists. These profile plots are used by the proposed WeedOut to detect similar crops and cluster them.

Now that profile plots of every crop in the image are known, each profile plot is compared with every other profile plot using Dynamic Time Warping (DTW) [27]. DTW is an algorithm used to find the similarity between two time series which vary in time and speed. DTW stretches and compresses one of the signals along the time axis as represented

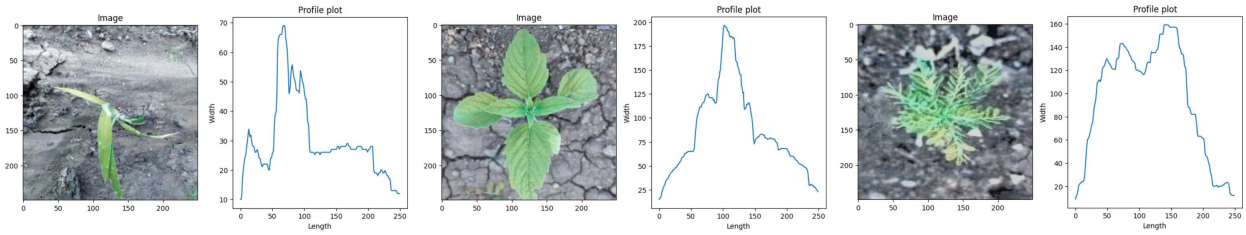


Figure 5: Visualization of profile plots.

in Figure 6 to compare them. This process of manipulating a signal on a time axis is termed time warping. DTW finds optimal alignment between the sequences based on the least cumulative distance rather than exactly matching data points like in Euclidean distance. Apart from finding the optimal alignment, DTW also produces the DTW distance which is the cost of aligning the two signals as defined in Algorithm **??**. In simple terms, DTW produces higher distances for signals that are dissimilar and lower distances for signals that are similar. So, DTW distance can be considered as a measure of similarity between two different time series signals. This capability has made DTW applicable for comparing different signals in various domains like speech detection [28], gait analysis [29], signature verification [30] and word matching [31].

Profile plot of a crop is essentially a graph where the number of columns occupied by the image is plotted on the Y axis against the row number on the X axis. By assuming the X axis as a time axis, it can be considered as time series data and DTW can be performed on them. Authors of [31] presented a method to match words by comparing their profile plots using DTW. Similarly, WeedOut tries to identify and cluster similar crops by comparing their profile plots using DTW.

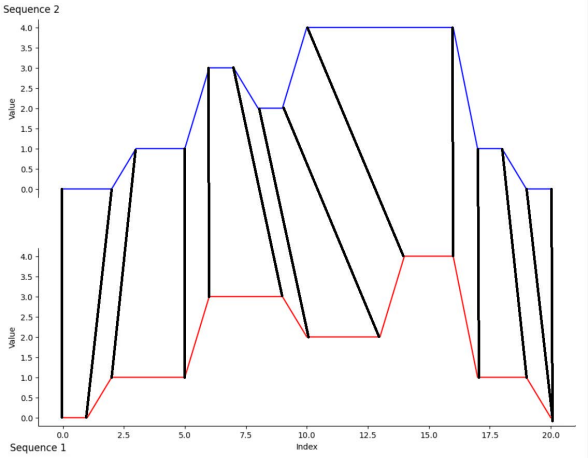


Figure 6: Illustration of Dynamic Time Warping.

When DTW distance is computed between all pairs of profile plots available, profile plots with the least DTW distance are considered to be similar. This helps in finding all pairs of crops that are similar to each other which are essentially the clusters with two crops in each. There could be a scenario where it can output two clusters A and B in which A has crop 1 and crop 2 where B has crop 2 and crop 3. From these clusters it can be inferred that crop 1 is similar to crop 2, crop 2 is similar to crop 3 and by property of association crop 1, crop 2 and crop 3 are similar. Thus crop 1, crop 2 and crop 3 can be added to a single clustered. To handle all such scenarios all the generated clusters are examined for any intersection between them in iteration till no intersection is found. While iterating through the clusters, if any two clusters having an intersection are found, they will be merged to form a single cluster similar to density-based clustering [32]. Algorithm **??** details the clustering algorithm adapted by WeedOut to output a minimum number of clusters of crops possible with given profile plots.

Comparing crops in images using DTW and clustering would produce a few clusters per image. Processing all the captured images similarly will produce a large number of clusters and it will be not feasible for the farmer to go through all the clusters and identify the clusters representing the primary crop. So, initially couple of images are combined to generate an image whose size is proportional to the number of images combined and the resolutions mentioned in 4.1.

This combined image is processed to extract crops, identify individual crops and cluster identical crops as discussed above. Since multiple images are considered, the images will contain samples of most of the crop types present in the farmland. Performing clustering on these images will result in few number of clusters representing all types of crops present in the farmland. Profile plots from each cluster are averaged to generate a single profile plot representing each cluster. Thereafter, profile plots generated for crops in each leftover images are compared with the profile plots of identified clusters. Each crop is then assigned to the cluster whose profile plot had the least DTW distance with the crop's profile plot.

Unlike CNN methods which do multiple passes over the image to look for various features in the image, the proposed method passes through the image only twice to identify crops and cluster them by DTW. Since these are non-CNN methods, the proposed method looks for no feature of leaves or crops and is not affected by changes in the appearance of crops due to geographical, biological, genetical and aging factors. After processing all the images, samples from each cluster are presented to the farmer.

4.4 Classify and Locate Weeds, Primary Crops

Once all possible clusters are identified, the area occupied by each crop in the image is identified by summing up the number of pixels that the crop object occupies in the image. The sum of the areas occupied by each crop in a cluster gives the area occupied by the cluster. The ratio of the area occupied by each cluster to the area occupied by the total vegetation gives the contribution of that cluster to the total vegetation. Since the images cannot relate to the actual physical measurements of crops, areas of crops computed by pixels have no real-time relevance. However, the pixel-based area can provide an accurate percentage of the area occupied by the crop. So, the percentage of vegetation would be a more meaningful feature than computed areas. All the identified clusters are presented to the farmer with a few images of crops from the cluster along with the percentage of contribution of that cluster to the total vegetation.

From the list of clusters presented to the farmer, he classifies/labels those clusters with images of his primary crop as the primary crop. All the crops that belong to the clusters identified by the farmer are classified as primary crops while all the crops from the rest of the clusters are classified as weeds. Thus labeling is performed in a semi-supervised fashion with minimal manual inputs required from the farmer to label a few instances after which the machine propagates the labels.

Given that each crop is classified as either primary crop or weed, the total area occupied by all primary crops and all weeds are computed to determine the percentage of contribution by weeds to the total vegetation by the Equation 1. This percentage is known as weed pressure and can provide insights on weed infestation to the farmer [33].

$$\text{Weed pressure} = \frac{\text{Area occupied by weed crops}}{\text{Total area occupied by all crops}} \tag{1}$$

For the autonomous weed sprayer to differentiate easily between weeds and primary crops [34], all weeds in the images are colored in red and primary crops are colored in green as shown in Figure 7. With the coloring of images, the classification task is considered to be completed and images are presented to weed sprayer for targeted spraying. As the proposed WeedOut utilizes geometric features and domain knowledge of the farmer the classification is robust and context-aware. The final results are presented to the farmer and saved on a cloud server for future needs.

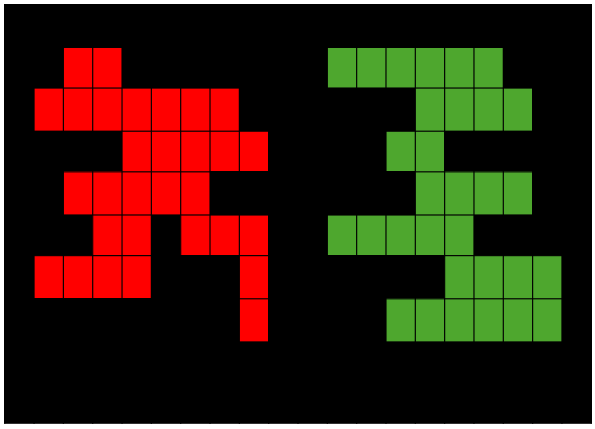


Figure 7: Colored image presented to the autonomous sprayer.

4.5 Eliminate Weeds

An autonomous weed sprayer [35] is a rover that can navigate through the field with provision to carry herbicide. These are used to spray herbicides at specific locations. The colored image generated by the algorithm is a minimized representation of a part of the farm. For instance, the image of 1200×2000 pixels could represent land of 3m×5m in real time. Initially, the weed sprayers will have to be calibrated to map the images to the size of the farmland. Once the calibration is performed, the weed sprayer starts to navigate along the farm in the path the rover/drone initially traversed to capture the images farm. The weed sprayer starts processing each pixel of the image column after column in each row. When the processor finds a red color pixel belonging to the identified weed, the spray nozzles are turned on to spray herbicide at the location. The nozzles remain turned off when a pixel of other colors is encountered. By the end of processing the images, it would spray herbicide at all locations colored red, eventually spraying herbicide on all the weeds.

5 Experimental Results

The solution proposed by WeedOut has been implemented using Python and a Computer Vision library OpenCV. The crop and weed data set from Kaggle [36] had individual images of various crops in the farmland. To create a data set representing farmland, multiple single crops (primary crops and weed crops) have been combined to form a single image with various crops. Results produced by WeedOut on these images created are presented in this section.

Initially to gather all kinds of crops present in the farmland, a few images from the generated data set are once again combined. Figure 8 shows the combined image produced. Thresholding is performed on the inputted image to eliminate the background soil and extract the crops. The color image is then converted to a binary image representing the background with a "0" and crops by "1". The binary image is next presented to the Two-Pass Connected Component Labeling Algorithm. Connected Component Labeling Algorithm identified 29 different objects in the binary image and labeled them from 1 to 29. Since converted the image is displayed as a gray image each identified crop has been shown in a different shade of gray in Figure 8. As there are many objects, when the image was presented in grayscale few crops are not clearly visible in the image. For all the 29 crops identified in Figure 8, profile plots are generated by counting instances of each label in all the rows where the label exists. Figure 9 shows all the profile plots generated.

Dynamic Time Warping (DTW) is then performed to compare all 29 crops with each other crop and to identify pairs that are similar by their DTW distance. All the identified pairs are again merged (union operation) to 7 clusters shown in Figure10. DTW clustered the 29 crops into 7 different clusters and assigned labels 1, 23, 27, 29, 13, 25,28. As the proposed algorithm tries to identify the crop with the least DTW distance as similar crops, a few crops did not have the same type of crop and were clustered with the closest type of crop in the image like the grass crop from the cluster 29. So, there have been some misclassifications while clustering.

Profile plots of all crops with the same labels were averaged so that they represent profile plots of all crops in the cluster. Figure 11, shows 7 profile plots for each of 7 clusters in Figure10. Areas of all crops in each cluster are computed by summing up the number of instances of labels of each crop that belongs to the cluster.

As of now, all possible clusters (set of similar crops) in the simulated farmland are formed and the rest of the images are processed individually. Results of two different images processed individually are presented in Figure 12. Image 1 had 6 different crops in the image. After identifying each crop, their profile plots are compared with profile plots in Figure11. As a result, they were assigned to clusters 17, 28, 28, 27, 29, 29 respectively and the areas of each cluster were updated. Similarly, for Image 2, the crops were assigned to clusters 29, 27, 13, 27, 13, 13. The same process is repeated for the rest of the images in the data set created to complete the clustering.

For the farmer to identify the cluster representing primary crops, he has to be presented with the images of crops in the clusters. Visualizing all the crops in all clusters needs more resources and also is hard for farmers to review all of them. Since the clusters present in the image Figure10 were the primary clusters, they could represent the entire cluster and are presented to the farmer. In this experiment, from the clusters presented, farmer identified the clusters 27 and 23 as clusters representing his primary crop. With this, all the crops present in clusters 27 and 23 are classified as primary crops and the rest are classified as weeds completing the semi-supervised classification.

The total area occupied by all crops and weed crops has been computed by summing areas of all crops of the same class as shown in Figure 13. For the presented data set, the weed pressure computed is 52%. The binary images produced while processing the images are used to color all the crops that were identified as weeds in red and all crops identified as primary crops in green. Figure 14 represents the colored image for Image 2 discussed earlier. Colored images and results will be saved on a cloud server for future use, these images will be fed to an autonomous weed sprayer to spray herbicide at locations colored red so that weeds are killed.

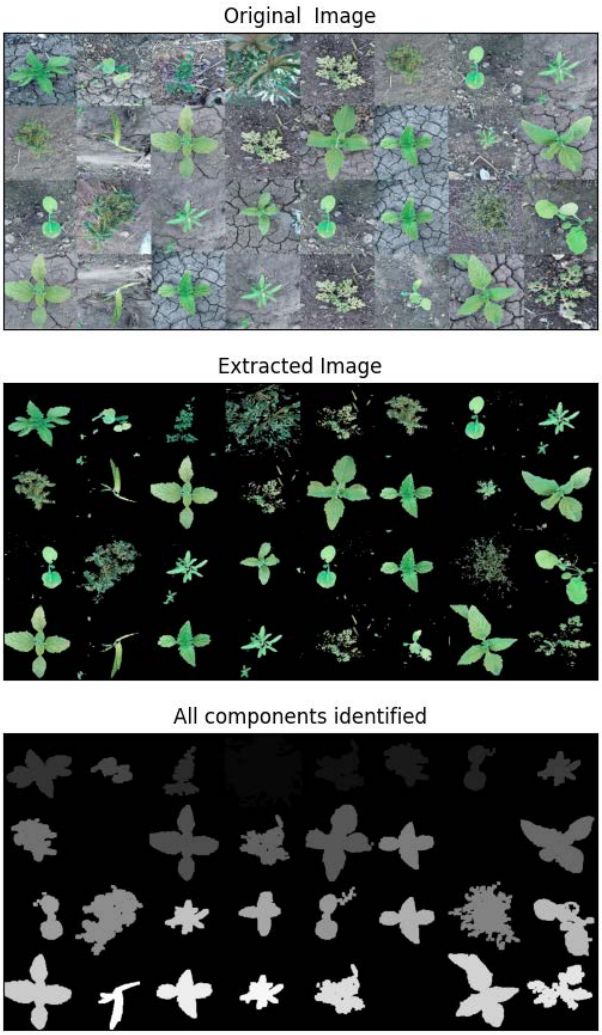


Figure 8: Different stages in crop identification.

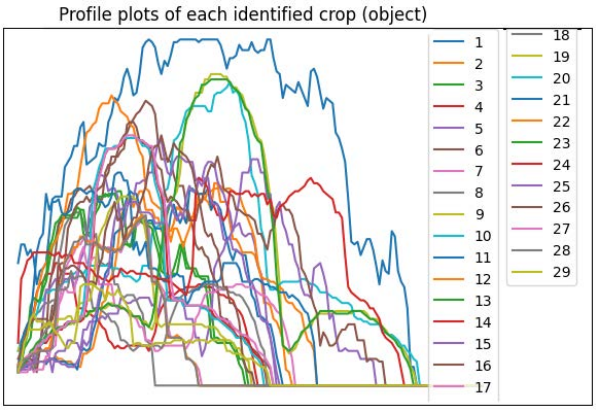


Figure 9: Profile plots for all crops in the field.

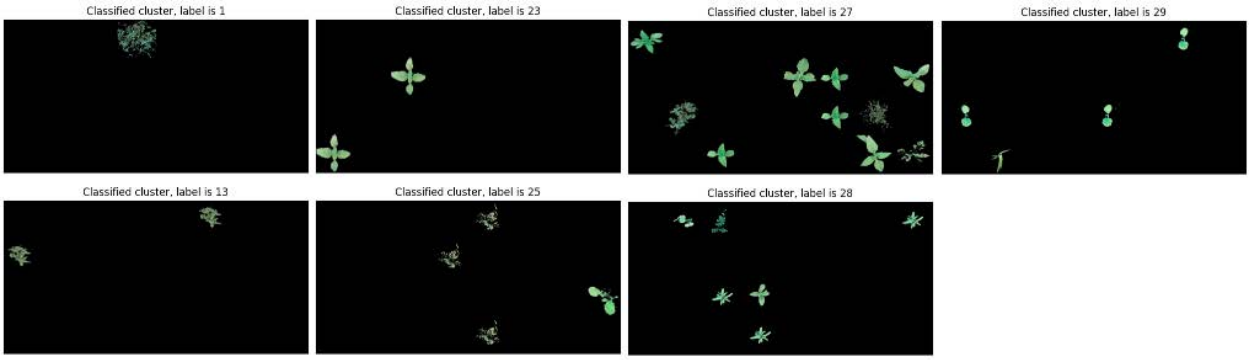


Figure 10: Clusters of similar crops.

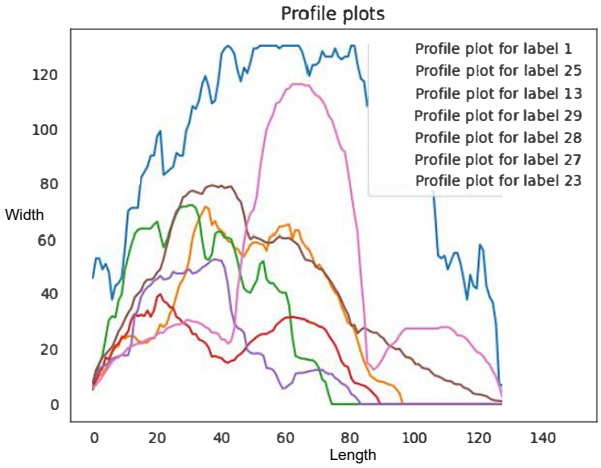
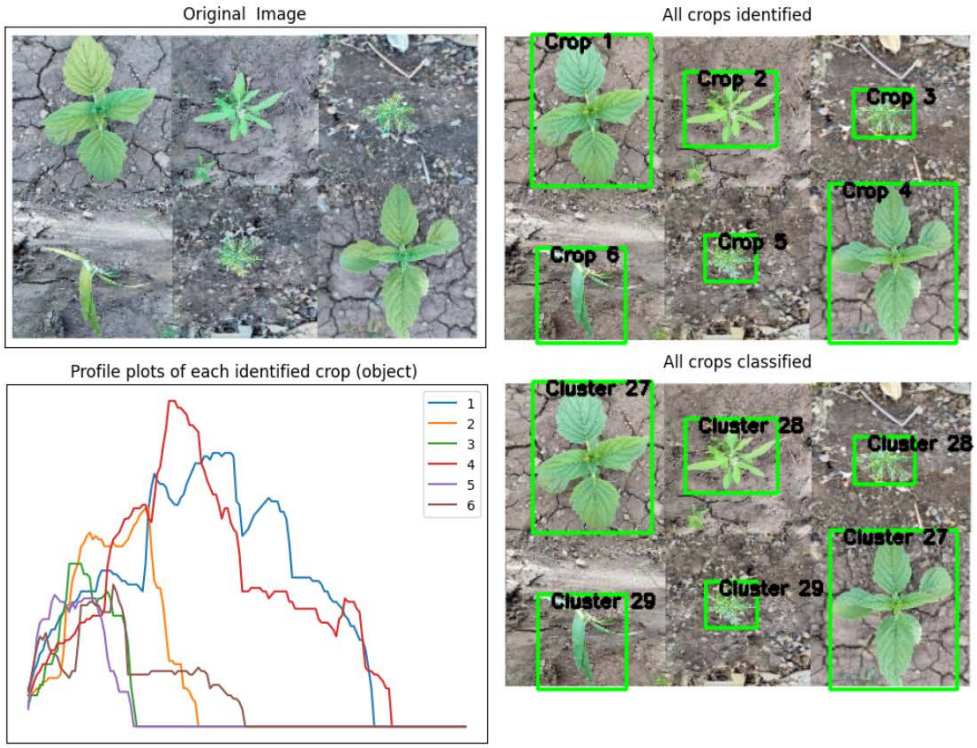


Figure 11: Profile plots of clusters of similar crops.

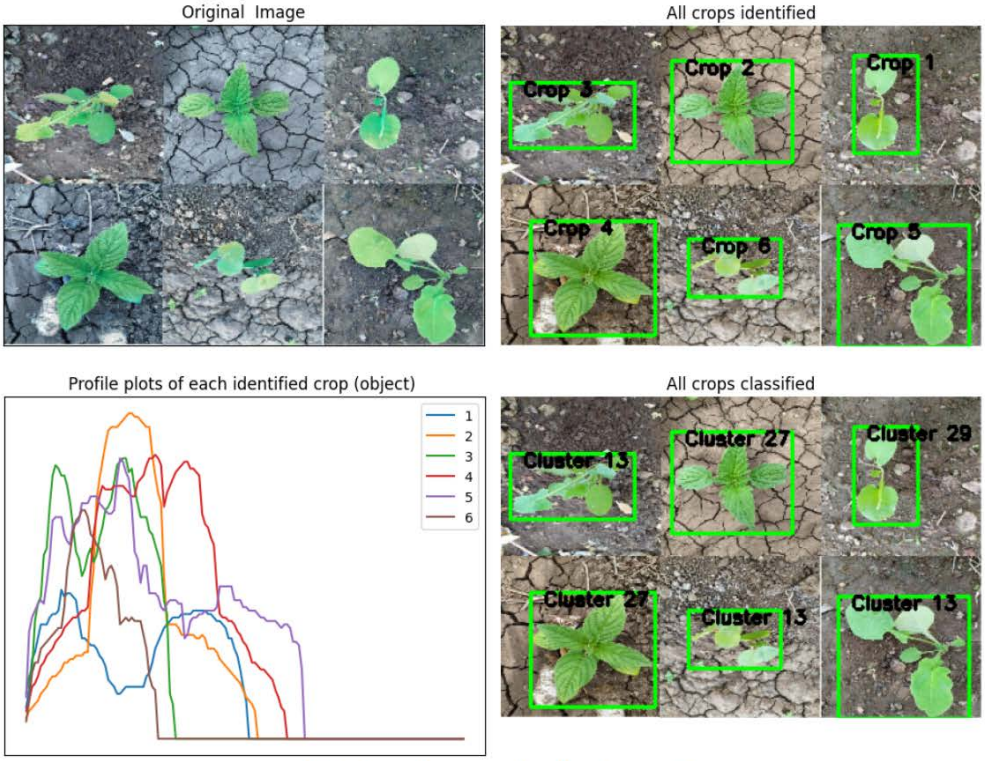
The presented clustering method was able to produce results similar to ground truth in terms of identifying primary crops and weeds with minimal human inputs. The data set generated to simulate the farmland had 20 images with a total of 120 individual crops of 8 different types. All these images were processed as per the method discussed and its performance metrics were calculated. Since we use profile plots and DTW to cluster similar crops if there is only one instance of any type of crop or if there are two kinds of crops whose shapes and area occupied are similar, they may be clustered together resulting in misclassifications affecting the accuracy of the model. Table **??** presents the confusion matrix for the proposed clustering, which achieved an accuracy of 80%.

From the presented 8 different types of crops, WeedOut formed 7 clusters. As Cluster 23 and 27 represent the primary crop, the actual number of different clusters identified is 6. Since class 1 represents primary crops, all the other classes are combined to form a weed class and the updated confusion matrix is presented in Table **??**.

Metrics computed for the proposed Non-CNN, Semi-Supervised classification model are presented in Table **??**. The method showed an accuracy of 89% demonstrating that when it classified 100 crops as weeds and primary crops, it was able to correctly classify 89 crops. Precision, the accuracy of the model in classifying primary crops is 82% while Recall, the percentage of primary crops it was able to classify correctly is 88%. F1 score, the harmonic mean of precision and recall of classification performed is 86%. This indicates that the proposed model can correctly identify instances of primary crops while minimizing the misclassification of primary crops as weeds. The wrong classification rate for primary crops (False Negative Rate) is 11% and 10% for weeds (False Positive Rate). Though WeedOut performed better in identifying and classifying weeds, it suffers from a few limitations which will be discussed in Section 6.



(a) Clustering results for Image 1



(b) Clustering results for Image 2

Figure 12: Results for 2 different images from the data set.

```
primary_crops ={27,23}
total_area = 0
weed_area = 0
for cluster, area in areas.items():
    total_area = total_area+area
    if cluster in primary_crops:
        weed_area = weed_area + area

weed_pressure = weed_area/total_area
print(f"Computed weed pressure is {weed_pressure}")
```

```
Computed weed pressure is 0.5264900224659707
```

Figure 13: Computed weed pressure.

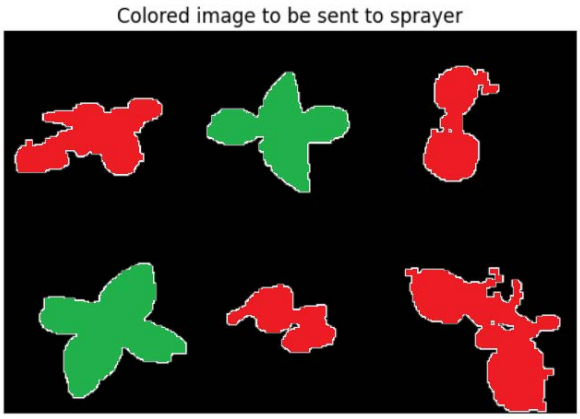


Figure 14: Output image highlighting weeds in red and primary crop in green.

6 Conclusion

Farmers will have to implement new technologies like smart agriculture to best utilize resources and maximize yield to cater to the rising population which is expected to reach 10 billion by 2050. Many solutions and advancements are also needed in weed management to make sure primary crops do not suffer from weeds. In this paper, we presented WeedOut, a disease management A-CPS in smart agriculture that relies on the shape, size of the crop and domain knowledge of the farmer in a semi-supervised method to detect weeds in the farmland. The presented method does not utilize any environment-dependent features as in CNN-based methods making it suitable for a wide range of crops and geographic locations. WeedOut uses the Two-Pass Connected Component algorithm to identify crops uniquely. It detects objects in the image by passing the same label to all the pixels which are connected. In cases where there is an overlap between crops in the image, the binary image will have one big component and the connected component algorithm process will identify it as one crop instead of two. While classification, such wrongly identified objects will be misclassified affecting the overall efficiency of the model. For the future developments of WeedOut, including methods to efficiently differentiate and individually identify each crop, even if they overlap, such as edge detection and including other geometric features in addition to the shape, size of the crop can be considered to improve the robustness and accuracy of the model.

Compliance with Ethical Standards

The authors declare that they have no conflict of interest and there was no human or animal testing or participation involved in this research. All data were obtained from public domain sources.

Acknowledgment

This article is an extended version of our previous conference paper presented at [37]. Authors would like to acknowledge the help of Dr. Alakananda Mitra for the help on the conference version of this work.

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